Supplementary Appendix E - Full Statistical Code

1. Packages

```
required_packages <- c("mada", "meta", "metafor", "dplyr", "ggplot2",
"writexl")
for (pkg in required_packages)
  if (!require(pkg, character.only = TRUE)) {
    install.packages(pkg)
    library(pkg, character.only = TRUE)
```

2. Package Version & Session Info

```
packageVersion("mada")
packageVersion("meta")
packageVersion("metafor")
packageVersion("dplyr")
packageVersion("ggplot2")
packageVersion("writexl")

sessionInfo()
```

```
packageVersion("mada")
[1] '0.5.12'
> packageVersion("meta")
[1] '8.3.0'
> packageVersion("metafor")
[1] '4.8.0'
> packageVersion("dplyr")
[1] '1.2.0'
> packageVersion("ggplot2")
[1] '4.0.3'
> packageVersion("writexl")
[1] '1.5.4'
>
> sessionInfo()
R version 4.5.3 (2026-03-11 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26200)

Matrix products: default
  LAPACK version 3.12.1
```

```
locale:
[1] LC_COLLATE=English_New Zealand.utf8
[2] LC_CTYPE=English_New Zealand.utf8
[3] LC_MONETARY=English_New Zealand.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=English_New Zealand.utf8

time zone: Europe/London
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods
[7] base

other attached packages:
 [1] writexl_1.5.4       mada_0.5.12         metafor_4.8-0
 [4] numDeriv_2016.8-1.1 Matrix_1.7-4        mvmeta_1.0.3
 [7] ellipse_0.5.0       mvtnorm_1.3-6       ggplot2_4.0.3
[10] dplyr_1.2.0         meta_8.3-0          metadat_1.4-0
[13] metabook_0.2-0      devEMF_4.5-1

loaded via a namespace (and not attached):
 [1] mixmeta_1.2.2      generics_0.1.4     xml2_1.5.2
 [4] stringi_1.8.7      lattice_0.22-9     lme4_2.0-1
 [7] hms_1.1.4          digest_0.6.39      magrittr_2.0.4
[10] grid_4.5.3         RColorBrewer_1.1-3 CompQuadForm_1.4.4
[13] purrr_1.2.1        scales_1.4.0       reformulas_0.4.4
[16] Rdpack_2.6.6       cli_3.6.5          rlang_1.1.7
[19] rbibutils_2.4.1    splines_4.5.3      withr_3.0.2
[22] tools_4.5.3        tzdb_0.5.0         nloptr_2.2.1
[25] minqa_1.2.8        mathjaxr_2.0-0     boot_1.3-32
[28] vctrs_0.7.1        R6_2.6.1           lifecycle_1.0.5
[31] stringr_1.6.0      MASS_7.3-65        pkgconfig_2.0.3
[34] pillar_1.11.1      gtable_0.3.6       glue_1.8.0
[37] Rcpp_1.1.1         tibble_3.3.1       tidyselect_1.2.1
[40] farver_2.1.2       nlme_3.1-168       labeling_0.4.3
[43] readr_2.2.0        compiler_4.5.3     S7_0.2.1
```

3. Hardcoded extracted 2x2 data

```
data <- data.frame(
  StudyID = c("AbuAlrub2022","AbuAlrub2022","AbuAlrub2022",

"Mannhardt2023","Mannhardt2023","Mannhardt2023","Mannhardt2023",

"Muller2024","Briosa2025","OscaAsensi2021","Wouters2025","Racine2022"
),
  Device = c("Apple","Samsung","Withings",
             "Apple","Fitbit","Samsung","Withings",
```

```
              "Apple","Apple","RITHMI","Apple","Apple"),
  TP = c(87,88,78,52,52,35,40,13,134,92,30,120),
  FN = c(13,12,22,9,9,26,21,5,60,6,0,34),
  FP = c(14,19,20,35,35,35,29,6,79,1,2,98),
  TN = c(86,81,80,105,105,105,111,69,211,34,90,482)
)
```

Script 1. Bivariate Meta-Analysis (Reitsma Model)

Key Results:

- Sensitivity: 0.813 (95% CI: 0.735–0.872)
- AUC: 0.885

Interpretation: Indicates good overall diagnostic accuracy, with the test correctly identifying most AF cases and non-cases, though not sufficient as a standalone diagnostic tool.

```
# Step 1: Install package (only once)
install.packages("mada")
# Step 2: Load package
library(mada)
# Step 3: Create your dataset
data <- data.frame(
  StudyID = c("AbuAlrub2022","AbuAlrub2022","AbuAlrub2022",
              "Mannhardt2023","Mannhardt2023","Mannhardt2023","Mannhardt2023",
              "Muller2024","BrisessionInfo()
osa2025","OscaAsensi2021","Wouters2025","Racine2022"),
  Device = c("Apple","Samsung","Withings",
             "Apple","Fitbit","Samsung","Withings",
             "Apple","Apple","RITHMI","Apple","Apple"),
  TP = c(87,88,78,52,52,35,40,13,134,92,30,120),
  FN = c(13,12,22,9,9,26,21,5,60,6,0,34),
  FP = c(14,19,20,35,35,35,29,6,79,1,2,98),
  TN = c(86,81,80,105,105,105,111,69,211,34,90,482)
)
# Step 4: Run the model
model <- reitsma(data)
# Step 5: PRINT RESULTS
summary(model)
packageVersion("mada")
fpr <- 0.175
fpr_lower <- 0.133
fpr_upper <- 0.227
specificity <- 1 - fpr
spec_lower <- 1 - fpr_upper
```

```
spec_upper <- 1 - fpr_lower
specificity
spec_lower
spec_upper
```

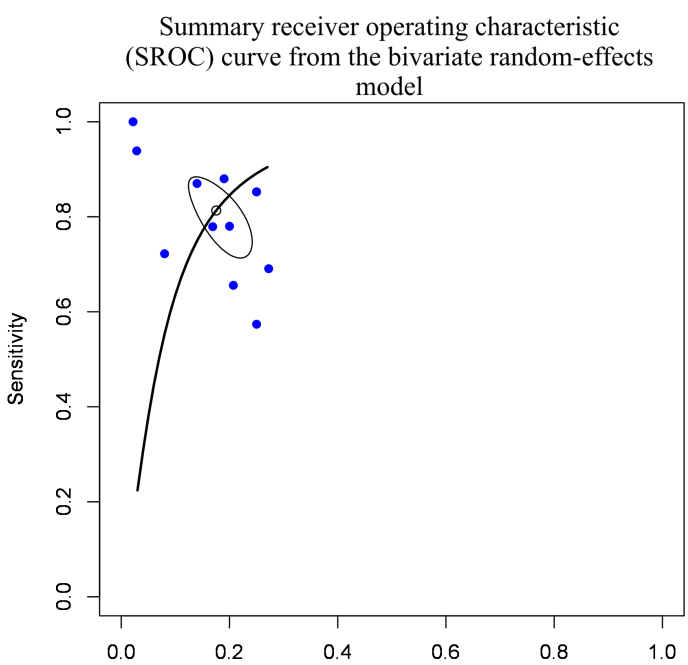
```
Call:  reitsma.default(data = data)

Bivariate diagnostic random-effects meta-analysis
Estimation method: REML

Fixed-effects coefficients
                  Estimate Std. Error      z Pr(>|z|) 95%ci.lb
tsens.(Intercept)    1.471      0.229  6.427    0.000    1.023
tfpr.(Intercept)    -1.548      0.166 -9.333    0.000   -1.873
sensitivity          0.813          -      -        -    0.735
false pos. rate      0.175          -      -        -    0.133
                  95%ci.ub
tsens.(Intercept)    1.920 ***
tfpr.(Intercept)    -1.223 ***
sensitivity          0.872
false pos. rate      0.227
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Variance components: between-studies Std. Dev and correlation matrix
      Std. Dev  tsens  tfpr
tsens    0.711  1.000     .
tfpr     0.506 -0.831 1.000
logLik     AIC     BIC
 25.333 -40.666 -34.776

AUC:  0.885
Partial AUC (restricted to observed FPRs and normalized):  0.685

I2 estimates
Zhou and Dendukuri approach:  20 %
Holling sample size unadjusted approaches:  61.3 - 85.9 %
Holling sample size adjusted approaches:  3.2 - 5.6 %>
```

```
packageVersion("mada")
[1] '0.5.12'
> fpr <- 0.175
> fpr_lower <- 0.133
> fpr_upper <- 0.227
> specificity <- 1 - fpr
> spec_lower <- 1 - fpr_upper
> spec_upper <- 1 - fpr_lower
> specificity
[1] 0.825
```

```
> spec_lower
[1] 0.773
> spec_upper
[1] 0.867
```

SROC PLOT

```
library(devEMF)
>
> # Open EMF device
> emf("SROC_plot.emf", width = 7, height = 7)
>
> # Fix margins (prevents "figure margins too large")
> par(mar = c(4, 4, 3, 1))
>
> # Plot SROC curve
> plot(model, sroclwd = 2)
>
> # Add study points
> points(
+     1 - (data$TN / (data$TN + data$FP)),  # FPR
+     data$TP / (data$TP + data$FN),        # Sensitivity
+     pch = 16,
+     col = "blue"
+ )
>
> # Close device
> dev.off()
```



Script 2: HSROC Model

Key Results:

- Sensitivity: 0.845 (95% CI: 0.741–0.912)
- Specificity: 0.866
- AUC: 0.919

Interpretation: Shows slightly improved performance when accounting for study-level clustering, suggesting robust discriminative ability across studies.

```
library(mada)
library(dplyr)
packageVersion("mada")
packageVersion("dplyr")
data <- data.frame(
StudyID = c("AbuAlrub2022","AbuAlrub2022","AbuAlrub2022",
"Mannhardt2023","Mannhardt2023","Mannhardt2023","Mannhardt2023",
"Muller2024","Briosa2025","OscaAsensi2021","Wouters2025","Racine2022"),
TP = c(87,88,78,52,52,35,40,13,134,92,30,120),
FN = c(13,12,22,9,9,26,21,5,60,6,0,34),
FP = c(14,19,20,35,35,35,29,6,79,1,2,98),
TN = c(86,81,80,105,105,105,111,69,211,34,90,482)
)
data_study <- data %>%
group_by(StudyID) %>%
summarise(
TP = sum(TP),
FN = sum(FN),
FP = sum(FP),
TN = sum(TN),
.groups = "drop"
)
# Apply 0.5 continuity correction after study-level aggregation
data_study_cc <- data_study
data_study_cc$TP <- data_study_cc$TP + 0.5
data_study_cc$FN <- data_study_cc$FN + 0.5
data_study_cc$FP <- data_study_cc$FP + 0.5
data_study_cc$TN <- data_study_cc$TN + 0.5
# Create mada matrix using corrected data
mada_matrix <- as.matrix(data_study_cc[, c("TP", "FN", "FP", "TN")])
# Fit HSROC/Reitsma model
hsroc_model <- reitsma(mada_matrix)
summary(hsroc_model)
plot(hsroc_model)
png("hsroc_plot.png", width = 1200, height = 1200, res = 150)
```

```
par(mar = c(4, 4, 3, 1))
plot(hsroc_model, sroclwd = 2)
points(
1 - (data_study$TN / (data_study$TN + data_study$FP)),
data_study$TP / (data_study$TP + data_study$FN),
pch = 16,
col = "blue"
)
dev.off()
```

```
Call:  reitsma.default(data = mada_matrix)

Bivariate diagnostic random-effects meta-analysis
Estimation method: REML

Fixed-effects coefficients
                  Estimate Std. Error      z Pr(>|z|) 95%ci.lb
tsens.(Intercept)    1.693      0.327  5.181    0.000    1.053
tfpr.(Intercept)    -1.865      0.330 -5.658    0.000   -2.511
sensitivity          0.845          -      -        -    0.741
false pos. rate      0.134          -      -        -    0.075
                  95%ci.ub
tsens.(Intercept)    2.334 ***
tfpr.(Intercept)    -1.219 ***
sensitivity          0.912
false pos. rate      0.228
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Variance components: between-studies Std. Dev and correlation matrix
      Std. Dev  tsens  tfpr
tsens    0.807  1.000     .
tfpr     0.821 -1.000 1.000

 logLik     AIC     BIC
 18.201 -26.401 -23.206

AUC:  0.919
Partial AUC (restricted to observed FPRs and normalized):  0.808

I2 estimates
Zhou and Dendukuri approach:  0 %
Holling sample size unadjusted approaches:  49 - 90 %
Holling sample size adjusted approaches:  2.1 - 4.5 %
```

Specificity Calculation

```
# Look at coefficient structure
coef(hsroc_model)
# Extract values by column position/name
sens_logit <- coef(hsroc_model)[1, "tsens"]
fpr_logit  <- coef(hsroc_model)[1, "tfpr"]
# Convert from logit scale
sensitivity <- plogis(sens_logit)
fpr <- plogis(fpr_logit)
# Specificity
specificity <- 1 - fpr
sensitivity
Specificity
```

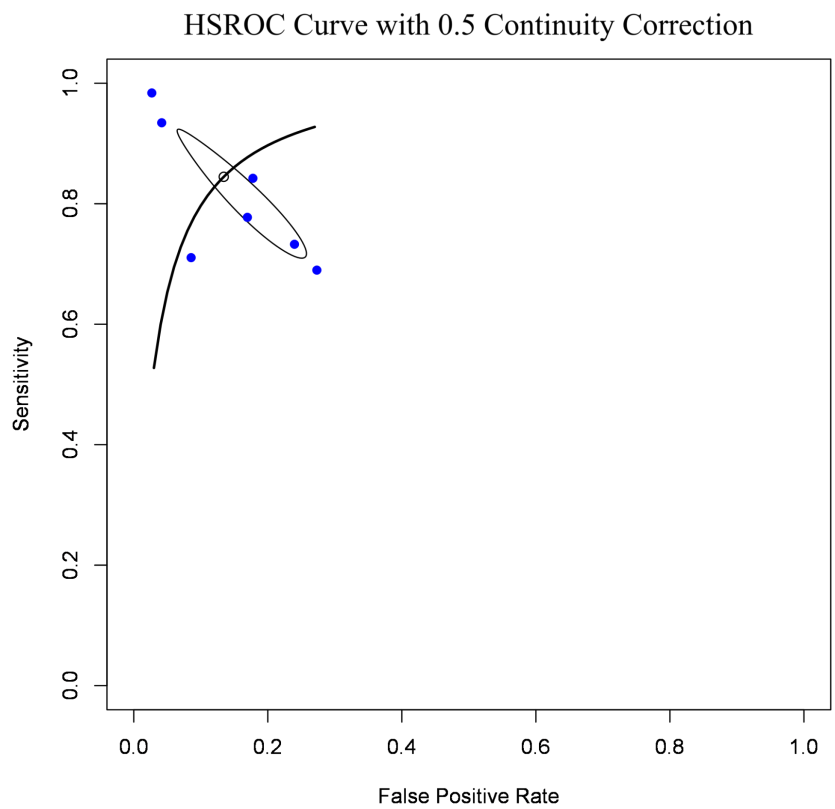
```
> sensitivity
[1] 0.8446802
> specificity
[1] 0.8658345
```

95% CI Calculation

```
s <- summary(hsroc_model)
# Sensitivity CI directly from summary
sens <- s$coefficients["sensitivity", "Estimate"]
sens_lower <- s$coefficients["sensitivity", "95%ci.lb"]
sens_upper <- s$coefficients["sensitivity", "95%ci.ub"]
# FPR CI from summary
fpr <- s$coefficients["false pos. rate", "Estimate"]
fpr_lower <- s$coefficients["false pos. rate", "95%ci.lb"]
fpr_upper <- s$coefficients["false pos. rate", "95%ci.ub"]
# Convert FPR to specificity
specificity <- 1 - fpr
spec_lower <- 1 - fpr_upper
spec_upper <- 1 - fpr_lower
sens
sens_lower
sens_upper
specificity
spec_lower
Spec_upper
```

```
> sens
[1] 0.8446802
> sens_lower
[1] 0.7413283
> sens_upper
[1] 0.9116589
>
> specificity
[1] 0.8658345
```

```
> spec_lower
[1] 0.7718301
> spec_upper
[1] 0.9248792
```



Script 3; Diagnostic Odds Ratio (DOR)

Key Results:

- Random-effects DOR: 17.97 (95% CI: 10.49–30.77)
- I^2: 84.0%

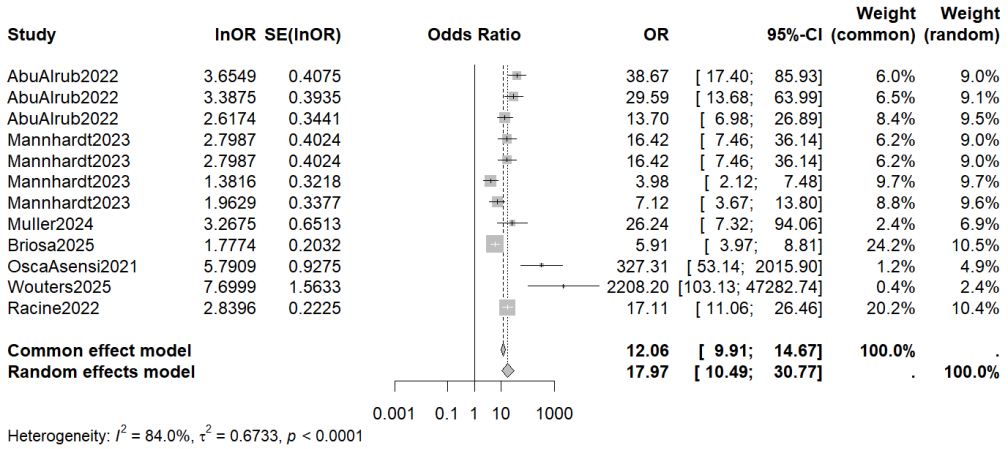
Interpretation:Demonstrates strong overall discrimination, but high heterogeneity indicates variability in performance across studies.

```
# Install packages if needed
install.packages("meta")
# Load libraries
library(meta)
# Create dataset
data <- data.frame(
```

```
StudyID = c("AbuAlrub2022","AbuAlrub2022","AbuAlrub2022",
"Mannhardt2023","Mannhardt2023","Mannhardt2023","Mannhardt2023",
"Muller2024","Briosa2025","OscaAsensi2021","Wouters2025","Racine2022"),
Device = c("Apple","Samsung","Withings",
"Apple","Fitbit","Samsung","Withings",
"Apple","Apple","RITHMI","Apple","Apple"),
TP = c(87,88,78,52,52,35,40,13,134,92,30,120),
FN = c(13,12,22,9,9,26,21,5,60,6,0,34),
FP = c(14,19,20,35,35,35,29,6,79,1,2,98),
TN = c(86,81,80,105,105,105,111,69,211,34,90,482)
)
# Apply continuity correction of 0.5 to all cells
data_cc <- data
data_cc$TP <- data$TP + 0.5
data_cc$FN <- data$FN + 0.5
data_cc$FP <- data$FP + 0.5
data_cc$TN <- data$TN + 0.5
# Calculate DOR
data_cc$DOR <- (data_cc$TP * data_cc$TN) / (data_cc$FP * data_cc$FN)
# Log-transform DOR
data_cc$logDOR <- log(data_cc$DOR)
# Standard error of log(DOR)
data_cc$SE_logDOR <- sqrt(
(1 / data_cc$TP) +
(1 / data_cc$FN) +
(1 / data_cc$FP) +
(1 / data_cc$TN)
)
# Run random-effects meta-analysis using DerSimonian-Laird estimator
dor_meta <- metagen(
TE = data_cc$logDOR,
seTE = data_cc$SE_logDOR,
studlab = data_cc$StudyID,
sm = "OR",
method.tau = "DL"
)
# View results
summary(dor_meta)
# Optional forest plot
forest(dor_meta)
```

```
                      OR                 95%-CI %W(common) %W(random)
AbuAlrub2022     38.6654 [ 17.3975;    85.9327]        6.0        9.0
AbuAlrub2022     29.5908 [ 13.6835;    63.9903]        6.5        9.1
AbuAlrub2022     13.7003 [  6.9800;    26.8908]        8.4        9.5
Mannhardt2023    16.4233 [  7.4628;    36.1425]        6.2        9.0
```

```
Mannhardt2023    16.4233 [  7.4628;    36.1425]        6.2        9.0
Mannhardt2023     3.9811 [  2.1188;     7.4803]        9.7        9.7
Mannhardt2023     7.1198 [  3.6727;    13.8025]        8.8        9.6
Muller2024       26.2448 [  7.3231;    94.0563]        2.4        6.9
Briosa2025        5.9144 [  3.9718;     8.8071]       24.2       10.5
OscaAsensi2021  327.3077 [ 53.1426;  2015.9043]        1.2        4.9
Wouters2025    2208.2000 [103.1274; 47282.7409]        0.4        2.4
Racine2022       17.1092 [ 11.0619;    26.4623]       20.2       10.4

Number of studies: k = 12

                         OR             95%-CI     z  p-value
Common effect model  12.0579 [ 9.9124; 14.6679] 24.90 < 0.0001
Random effects model 17.9685 [10.4941; 30.7665] 10.53 < 0.0001

Quantifying heterogeneity (with 95%-CIs):
 tau^2 = 0.6733 [0.5200; 4.2466]; tau = 0.8205 [0.7211; 2.0607]
 I^2 = 84.0% [73.6%; 90.3%]; H = 2.50 [1.95; 3.22]

Test of heterogeneity:
     Q d.f.  p-value
 68.96   11 < 0.0001

Details of meta-analysis methods:
- Inverse variance method
- DerSimonian-Laird estimator for tau^2
- Jackson method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
```



Script 4: PPV Meta-Analisis

Key Results:

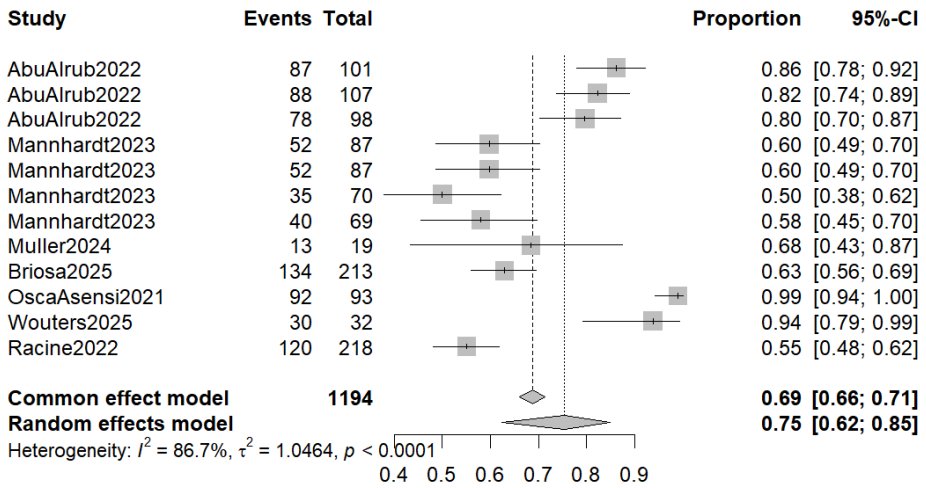
- PPV: 0.753 (95% CI: 0.623–0.849)

Interpretation: Suggests moderate rule-in ability; a notable proportion of positive results may be false positives, especially in low-prevalence settings.

```
install.packages("meta")
packageVersion("meta")
# Load library
library(meta)
# Create dataset
data <- data.frame(
StudyID = c("AbuAlrub2022","AbuAlrub2022","AbuAlrub2022",
"Mannhardt2023","Mannhardt2023","Mannhardt2023","Mannhardt2023",
"Muller2024","Briosa2025","OscaAsensi2021","Wouters2025","Racine2022"),
Device = c("Apple","Samsung","Withings",
"Apple","Fitbit","Samsung","Withings",
"Apple","Apple","RITHMI","Apple","Apple"),
TP = c(87,88,78,52,52,35,40,13,134,92,30,120),
FN = c(13,12,22,9,9,26,21,5,60,6,0,34),
FP = c(14,19,20,35,35,35,29,6,79,1,2,98),
TN = c(86,81,80,105,105,105,111,69,211,34,90,482)
)
# Calculate PPV per study/device
data$PPV <- data$TP / (data$TP + data$FP)
# Run random-effects meta-analysis of PPV
ppv_meta <- metaprop(
event = TP,
n = TP + FP,
studlab = StudyID,
data = data,
sm = "PLOGIT",
method.tau = "ML"
)
# View results
summary(ppv_meta)
# Optional forest plot
forest(ppv_meta)
```

```
               proportion           95%-CI
AbuAlrub2022       0.8614 [0.7784; 0.9221]
AbuAlrub2022       0.8224 [0.7367; 0.8896]
```

```
AbuAlrub2022        0.7959 [0.7026; 0.8707]
Mannhardt2023       0.5977 [0.4871; 0.7015]
Mannhardt2023       0.5977 [0.4871; 0.7015]
Mannhardt2023       0.5000 [0.3780; 0.6220]
Mannhardt2023       0.5797 [0.4548; 0.6976]
Muller2024          0.6842 [0.4345; 0.8742]
Briosa2025          0.6291 [0.5605; 0.6941]
OscaAsensi2021      0.9892 [0.9415; 0.9997]
Wouters2025         0.9375 [0.7919; 0.9923]
Racine2022          0.5505 [0.4818; 0.6177]

Number of studies: k = 12
Number of observations: o = 1194
Number of events: e = 821

                     proportion           95%-CI
Common effect model      0.6876 [0.6607; 0.7133]
Random effects model     0.7530 [0.6229; 0.8491]

Quantifying heterogeneity (with 95%-CIs):
 tau^2 = 1.0464; tau = 1.0229; I^2 = 86.7% [78.6%; 91.8%]; H = 2.74 [2.16;
3.48]

Test of heterogeneity:
           Q d.f.  p-value
 Wald  82.76   11 < 0.0001
 LRT  146.32   11 < 0.0001

Details of meta-analysis methods:
- Random intercept logistic regression model
- Maximum-likelihood estimator for tau^2
- Calculation of I^2 based on Q
- Logit transformation
- Clopper-Pearson confidence interval for individual studies
```



Script 5: NPV Meta-Analysis

Key Results:

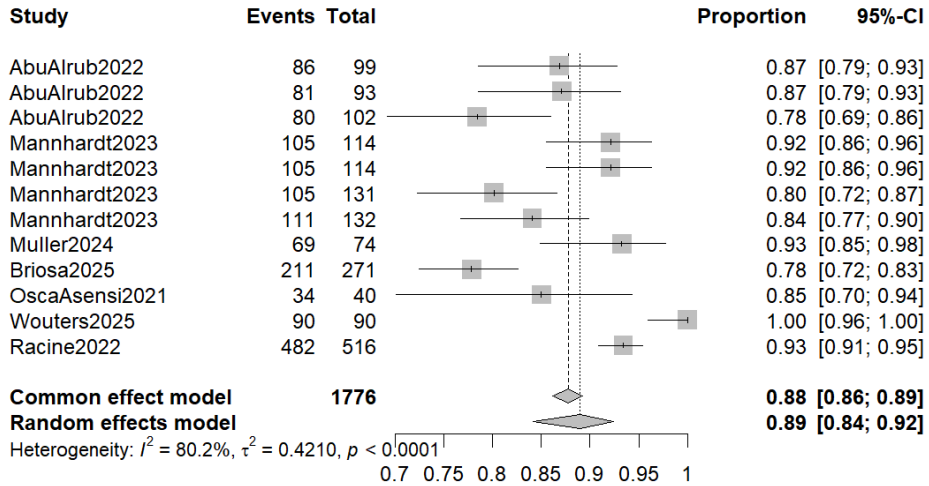
- NPV: 0.8897 [0.8417; 0.9244]

Interpretation: Indicates strong rule-out capability, meaning negative results are generally reliable.

```
# Load library
library(meta)
# Create dataset
data <- data.frame(
StudyID = c("AbuAlrub2022","AbuAlrub2022","AbuAlrub2022",
"Mannhardt2023","Mannhardt2023","Mannhardt2023","Mannhardt2023",
"Muller2024","Briosa2025","OscaAsensi2021","Wouters2025","Racine2022"),
Device = c("Apple","Samsung","Withings",
"Apple","Fitbit","Samsung","Withings",
"Apple","Apple","RITHMI","Apple","Apple"),
TP = c(87,88,78,52,52,35,40,13,134,92,30,120),
FN = c(13,12,22,9,9,26,21,5,60,6,0,34),
FP = c(14,19,20,35,35,35,29,6,79,1,2,98),
TN = c(86,81,80,105,105,105,111,69,211,34,90,482)
)
# Calculate NPV per study/device
data$NPV <- data$TN / (data$TN + data$FN)
# Run random-effects meta-analysis of NPV
npv_meta <- metaprop(
event = TN,
n = TN + FN,
studlab = StudyID,
data = data,
sm = "PLOGIT",
method.tau = "ML"
)
# View results
summary(npv_meta)
# Optional forest plot
forest(npv_meta)
```

```
               proportion           95%-CI
AbuAlrub2022       0.8687 [0.7859; 0.9282]
AbuAlrub2022       0.8710 [0.7855; 0.9315]
AbuAlrub2022       0.7843 [0.6919; 0.8596]
Mannhardt2023      0.9211 [0.8554; 0.9633]
```

```
Mannhardt2023      0.9211 [0.8554; 0.9633]
Mannhardt2023      0.8015 [0.7229; 0.8661]
Mannhardt2023      0.8409 [0.7672; 0.8987]
Muller2024         0.9324 [0.8493; 0.9777]
Briosa2025         0.7786 [0.7244; 0.8266]
OscaAsensi2021     0.8500 [0.7016; 0.9429]
Wouters2025        1.0000 [0.9598; 1.0000]
Racine2022         0.9341 [0.9091; 0.9539]

Number of studies: k = 12
Number of observations: o = 1776
Number of events: e = 1559

                     proportion           95%-CI
Common effect model      0.8778 [0.8617; 0.8922]
Random effects model     0.8897 [0.8417; 0.9244]

Quantifying heterogeneity (with 95%-CIs):
 tau^2 = 0.4210; tau = 0.6488; I^2 = 80.2% [66.2%; 88.4%]; H = 2.25 [1.72;
2.93]

Test of heterogeneity:
          Q d.f.  p-value
 Wald 55.53   11 < 0.0001
 LRT  84.03   11 < 0.0001

Details of meta-analysis methods:
- Random intercept logistic regression model
- Maximum-likelihood estimator for tau^2
- Calculation of I^2 based on Q
- Logit transformation
- Clopper-Pearson confidence interval for individual studies
```



Script 6: Multilevel Meta-Analysis (HSROC)

Key Results:

- Pooled DOR ≈ 30.12

Interpretation: Accounting for clustering increases estimated performance, suggesting within-study dependence influenced primary estimates.

```
install.packages("metafor")
# Load library
library(metafor)
packageVersion("metafor")
# Create dataset
data <- data.frame(
StudyID = c("AbuAlrub2022","AbuAlrub2022","AbuAlrub2022",
"Mannhardt2023","Mannhardt2023","Mannhardt2023","Mannhardt2023",
"Muller2024","Briosa2025","OscaAsensi2021","Wouters2025","Racine2022"),
Device = c("Apple","Samsung","Withings",
"Apple","Fitbit","Samsung","Withings",
"Apple","Apple","RITHMI","Apple","Apple"),
TP = c(87,88,78,52,52,35,40,13,134,92,30,120),
FN = c(13,12,22,9,9,26,21,5,60,6,0,34),
FP = c(14,19,20,35,35,35,29,6,79,1,2,98),
TN = c(86,81,80,105,105,105,111,69,211,34,90,482)
)
# Apply continuity correction
data_cc <- data
data_cc$TP <- data$TP + 0.5
data_cc$FN <- data$FN + 0.5
data_cc$FP <- data$FP + 0.5
data_cc$TN <- data$TN + 0.5
# Calculate DOR and log(DOR)
data_cc$DOR <- (data_cc$TP * data_cc$TN) / (data_cc$FP * data_cc$FN)
data_cc$logDOR <- log(data_cc$DOR)
# Calculate SE and variance of log(DOR)
data_cc$SE_logDOR <- sqrt(
(1 / data_cc$TP) +
(1 / data_cc$FN) +
(1 / data_cc$FP) +
(1 / data_cc$TN)
)
data_cc$vi <- data_cc$SE_logDOR^2
data_cc$yi <- data_cc$logDOR
# Run multilevel random-effects model
res_ml <- rma.mv(
```

```
yi = yi,
V = vi,
random = ~ 1 | StudyID/Device,
data = data_cc,
method = "REML"
)
# View model results
summary(res_ml)
# Back-transform pooled log(DOR) to DOR
exp(res_ml$b)
# Back-transform 95% CI to DOR scale
exp(c(res_ml$ci.lb, res_ml$ci.ub))
```

```
Multivariate Meta-Analysis Model (k = 12; method: REML)

  logLik  Deviance       AIC       BIC      AICc
-18.1856   36.3711   42.3711   43.5648   45.7997

Variance Components:

            estim    sqrt  nlvls  fixed          factor
sigma^2.1  1.9829  1.4081      7     no         StudyID
sigma^2.2  0.2950  0.5431     12     no  StudyID/Device

Test for Heterogeneity:
Q(df = 11) = 68.9603, p-val < .0001

Model Results:

estimate      se    zval    pval   ci.lb   ci.ub
  3.4051  0.6075  5.6050  <.0001  2.2144  4.5959  ***

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
            [,1]
intrcpt 30.11859
[1]  9.155914 99.077265
```

Script 7: Meta-Regression (Device Type)

Key Results:

- p = 0.515 (Not significant)

Interpretation:No statistically significant difference between devices, though analysis may be underpowered.

```
> install.packages("metafor")
```

```
> # Load libraries
> library(metafor)
> library(dplyr)
> # Create dataset
> data <- data.frame(
+    StudyID = c("AbuAlrub2022","AbuAlrub2022","AbuAlrub2022",
+
"Mannhardt2023","Mannhardt2023","Mannhardt2023","Mannhardt2023",
+
"Muller2024","Briosa2025","OscaAsensi2021","Wouters2025","Racine2022"),
+
+    Device = c("Apple","Samsung","Withings",
+               "Apple","Fitbit","Samsung","Withings",
+               "Apple","Apple","RITHMI","Apple","Apple"),
+
+    TP = c(87,88,78,52,52,35,40,13,134,92,30,120),
+    FN = c(13,12,22,9,9,26,21,5,60,6,0,34),
+    FP = c(14,19,20,35,35,35,29,6,79,1,2,98),
+    TN = c(86,81,80,105,105,105,111,69,211,34,90,482)
+ )
>
> # -------------------------------
> # Step 1: Remove devices used only once
> # -------------------------------
> data_meta <- data %>%
+    filter(Device != "RITHMI", Device != "Fitbit")
>
> # -------------------------------
> # Step 2: Continuity correction
> # -------------------------------
> data_meta <- data_meta %>%
+    mutate(
+        TP = ifelse(TP == 0, TP + 0.5, TP),
+        FP = ifelse(FP == 0, FP + 0.5, FP),
+        FN = ifelse(FN == 0, FN + 0.5, FN),
+        TN = ifelse(TN == 0, TN + 0.5, TN)
+    )
>
> # -------------------------------
> # Step 3: Calculate DOR
> # -------------------------------
> data_meta <- data_meta %>%
+    mutate(
+        DOR = (TP * TN) / (FP * FN),
+        logDOR = log(DOR),
+        var_logDOR = 1/TP + 1/FN + 1/FP + 1/TN
+    )
```

```
>
> # -------------------------------
> # Step 4: Run meta-regression
> # -------------------------------
> meta_model <- rma(
+    yi = logDOR,
+    vi = var_logDOR,
+    mods = ~ factor(Device),
+    method = "REML",
+    data = data_meta
+ )
>
> # -------------------------------
> # Step 5: View results
> # -------------------------------
> summary(meta_model)


Mixed-Effects Model (k = 10; tau^2 estimator: REML)

  logLik  deviance       AIC       BIC      AICc
-12.7704   25.5408   33.5408   33.3245   53.5408

tau^2 (estimated amount of residual heterogeneity):     1.0897 (SE = 0.7059)
tau (square root of estimated tau^2 value):             1.0439
I^2 (residual heterogeneity / unaccounted variability): 89.77%
H^2 (unaccounted variability / sampling variability):   9.78
R^2 (amount of heterogeneity accounted for):            0.00%

Test for Residual Heterogeneity:
QE(df = 7) = 54.8917, p-val < .0001

Test of Moderators (coefficients 2:3):
QM(df = 2) = 1.3262, p-val = 0.5153

Model Results:
                        estimate      se     zval    pval    ci.lb
intrcpt                   3.2003  0.4839   6.6133  <.0001   2.2518
factor(Device)Samsung    -0.8039  0.9192  -0.8746  0.3818  -2.6055
factor(Device)Withings   -0.8816  0.9156  -0.9629  0.3356  -2.6761
                         ci.ub
intrcpt                 4.1487  ***
factor(Device)Samsung   0.9976
factor(Device)Withings  0.9128

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Script 8: Publication Bias (Deeks Test)

Key Results:

- $p < 0.0001$

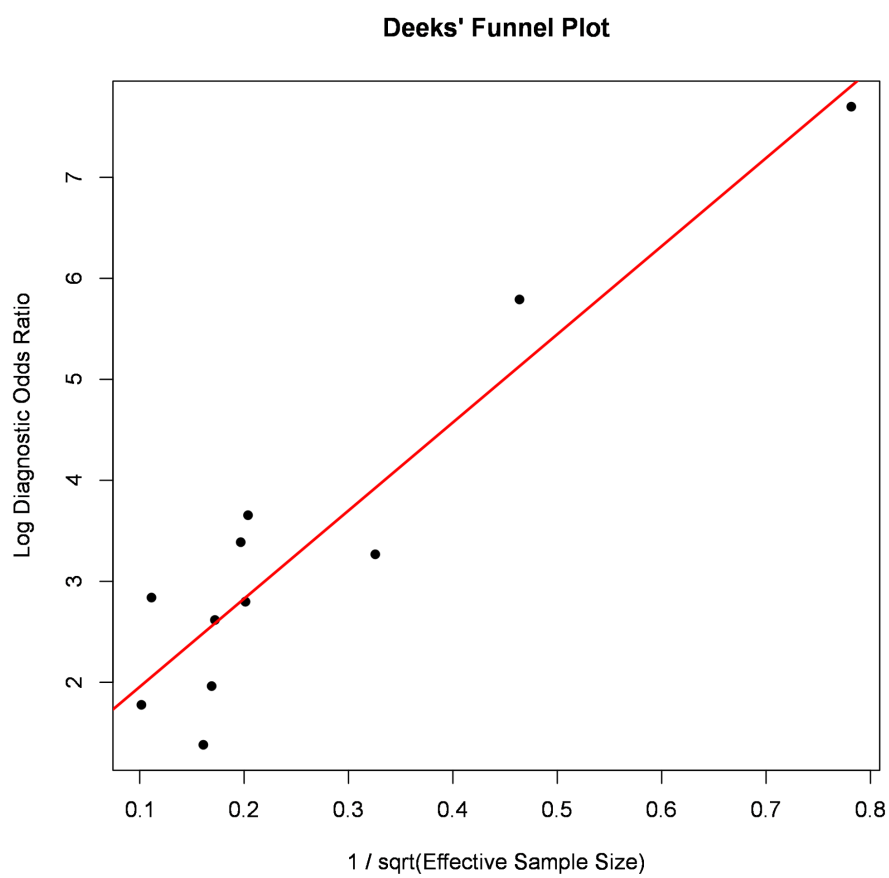
Interpretation: Suggests presence of small-study effects, though may also reflect heterogeneity rather than true publication bias.

```
library(metafor)
# Continuity correction
data_cc <- data
data_cc$TP <- data$TP + 0.5
data_cc$FN <- data$FN + 0.5
data_cc$FP <- data$FP + 0.5
data_cc$TN <- data$TN + 0.5
# DOR and log(DOR)
data_cc$DOR <- (data_cc$TP * data_cc$TN) / (data_cc$FP * data_cc$FN)
data_cc$logDOR <- log(data_cc$DOR)
# Standard error of log(DOR)
data_cc$SE_logDOR <- sqrt(
(1 / data_cc$TP) +
(1 / data_cc$FN) +
(1 / data_cc$FP) +
(1 / data_cc$TN)
)
# Effective sample size for Deeks test
data_cc$ESS <- 4 / (
(1 / data_cc$TP) +
(1 / data_cc$FN) +
(1 / data_cc$FP) +
(1 / data_cc$TN)
)
# Predictor for Deeks test
data_cc$inv_sqrt_ESS <- 1 / sqrt(data_cc$ESS)
# Deeks meta-regression
deeks_model <- rma(
yi = logDOR,
vi = SE_logDOR^2,
mods = ~ inv_sqrt_ESS,
method = "REML",
data = data_cc
)
summary(deeks_model)
# Predictor for Deeks test
data_cc$inv_sqrt_ESS <- 1 / sqrt(data_cc$ESS)
```

```r
# Deeks meta-regression
deeks_model <- rma(
yi = logDOR,
vi = SE_logDOR^2,
mods = ~ inv_sqrt_ESS,
method = "REML",
data = data_cc
)
summary(deeks_model)
# Deeks funnel plot
# Deeks funnel plot saved as EMF
library(devEMF)

dir.create("Outputs/Figures", recursive = TRUE, showWarnings = FALSE)

emf("Outputs/Figures/deeks_funnel_plot.emf", width = 7, height = 7)

par(mar = c(5, 5, 4, 2))

plot(
  data_cc$inv_sqrt_ESS,
  data_cc$logDOR,
  xlab = "1 / sqrt(Effective Sample Size)",
  ylab = "Log Diagnostic Odds Ratio",
  pch = 16,
  main = "Deeks' Funnel Plot"
)

abline(deeks_model, col = "red", lwd = 2)

dev.off()
```

```
Mixed-Effects Model (k = 12; tau^2 estimator: REML)

  logLik  deviance       AIC       BIC      AICc
-10.2350   20.4700   26.4700   27.3778   30.4700

tau^2 (estimated amount of residual heterogeneity):     0.2939 (SE = 0.2014)
tau (square root of estimated tau^2 value):             0.5421
I^2 (residual heterogeneity / unaccounted variability): 69.98%
H^2 (unaccounted variability / sampling variability):   3.33
R^2 (amount of heterogeneity accounted for):            72.39%

Test for Residual Heterogeneity:
QE(df = 10) = 35.6022, p-val < .0001
```

```
Test of Moderators (coefficient 2):
QM(df = 1) = 20.1497, p-val < .0001

Model Results:

              estimate      se    zval    pval   ci.lb    ci.ub
intrcpt         1.0831  0.4258  2.5440  0.0110  0.2487   1.9176
inv_sqrt_ESS    8.7259  1.9439  4.4888  <.0001  4.9159  12.5359

intrcpt         *
inv_sqrt_ESS  ***

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```



Script 9: Device Subgroup Analysis

Key Results:

- Apple: Sensitivity 0.820 | Specificity 0.855
- Samsung: Sensitivity 0.757 | Specificity 0.781
- Withings: Sensitivity 0.724 | Specificity 0.796

Interpretation: Differences observed between devices, but likely influenced by study-level factors rather than intrinsic device superiority.

```
# Function to extract estimates + CI from a reitsma model
extract_results <- function(model) {
s <- summary(model)
# Extract logit estimates
tsens <- coef(model)[1]
tfpr  <- coef(model)[2]
# Extract CI (logit scale)
tsens_lower <- s$coef["tsens.(Intercept)", "95%ci.lb"]
tsens_upper <- s$coef["tsens.(Intercept)", "95%ci.ub"]
tfpr_lower <- s$coef["tfpr.(Intercept)", "95%ci.lb"]
tfpr_upper <- s$coef["tfpr.(Intercept)", "95%ci.ub"]
# Convert to probability scale
sens <- plogis(tsens)
sens_lower <- plogis(tsens_lower)
sens_upper <- plogis(tsens_upper)
fpr <- plogis(tfpr)
spec <- 1 - fpr
spec_lower <- 1 - plogis(tfpr_upper)
spec_upper <- 1 - plogis(tfpr_lower)
return(c(sens, sens_lower, sens_upper, spec, spec_lower, spec_upper))
}
apple_vals    <- extract_results(model_apple)
samsung_vals  <- extract_results(model_samsung)
withings_vals <- extract_results(model_withings)
results_devices <- data.frame(
Device = c("Apple", "Samsung", "Withings"),
Sensitivity = c(apple_vals[1], samsung_vals[1], withings_vals[1]),
Sens_Lower  = c(apple_vals[2], samsung_vals[2], withings_vals[2]),
Sens_Upper  = c(apple_vals[3], samsung_vals[3], withings_vals[3]),
Specificity = c(apple_vals[4], samsung_vals[4], withings_vals[4]),
Spec_Lower  = c(apple_vals[5], samsung_vals[5], withings_vals[5]),
Spec_Upper  = c(apple_vals[6], samsung_vals[6], withings_vals[6])
)
print(results_devices)
results_table <- data.frame(
```

```
Device = results_devices$Device,
Sensitivity = sprintf(
"%.2f (%.2f-%.2f)",
results_devices$Sensitivity,
results_devices$Sens_Lower,
results_devices$Sens_Upper
),
Specificity = sprintf(
"%.2f (%.2f-%.2f)",
results_devices$Specificity,
results_devices$Spec_Lower,
results_devices$Spec_Upper
)
)
results_table
colnames(results_table) <- c(
"Device",
"Sensitivity (95% CI)",
"Specificity (95% CI)"
)
install.packages("writexl")
library(writexl)
write_xlsx(results_table, "device_results_table.xlsx")
```

```
 Device Sensitivity Sens_Lower Sens_Upper Specificity Spec_Lower
1    Apple   0.8196452  0.7270075  0.8857857   0.8547127  0.7532329
2  Samsung   0.7571637  0.3713614  0.9427171   0.7810670  0.6940671
3 Withings   0.7244182  0.5884136  0.8285744   0.7962360  0.7402128
  Spec_Upper
1  0.9189507
2  0.8487187
3  0.8427444
```

Device	Sensitivity	Specificity
1 Apple	**0.82 (0.73–0.89)**	**0.85 (0.75–0.92)**
2 Samsung	**0.76 (0.37–0.94)**	**0.78 (0.69–0.85)**
3 Withings	**0.72 (0.59–0.83)**	**0.80 (0.74–0.84)**

Bar Graph

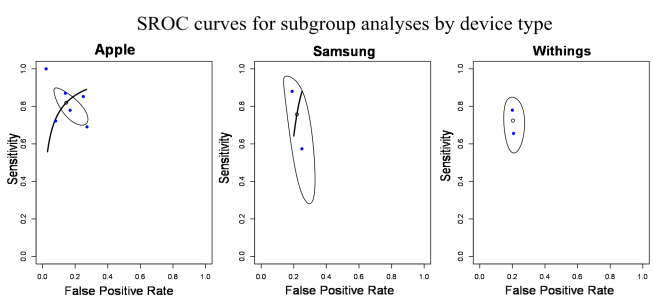
```
library(mada)
library(ggplot2)
data_apple <- subset(data, Device == "Apple")
data_samsung <- subset(data, Device == "Samsung")
data_withings <- subset(data, Device == "Withings")
model_apple <- reitsma(data_apple)
model_samsung <- reitsma(data_samsung)
```

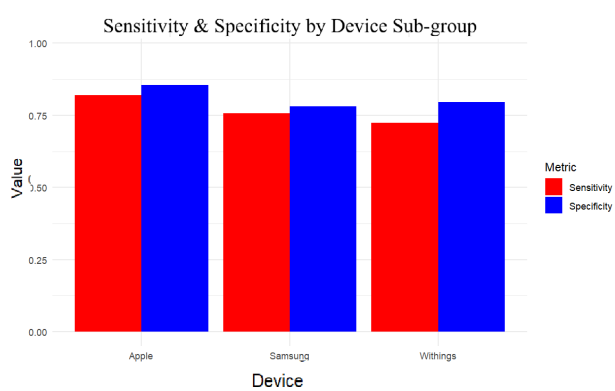
```
model_withings <- reitsma(data_withings)
# Apple
sens_apple <- plogis(coef(model_apple)[1])
spec_apple <- 1 - plogis(coef(model_apple)[2])
# Samsung
sens_samsung <- plogis(coef(model_samsung)[1])
spec_samsung <- 1 - plogis(coef(model_samsung)[2])
# Withings
sens_withings <- plogis(coef(model_withings)[1])
spec_withings <- 1 - plogis(coef(model_withings)[2])
sens_apple; spec_apple
results_devices <- data.frame(
Device = c("Apple", "Samsung", "Withings"),
Sensitivity = c(sens_apple, sens_samsung, sens_withings),
Specificity = c(spec_apple, spec_samsung, spec_withings)
)
results_devices
plot_data <- data.frame(
Device = rep(results_devices$Device, 2),
Metric = rep(c("Sensitivity", "Specificity"), each = 3),
Value = c(results_devices$Sensitivity, results_devices$Specificity)
)
ggplot(plot_data, aes(x = Device, y = Value, fill = Metric)) +
geom_bar(stat = "identity", position = "dodge") +
scale_fill_manual(values = c("Sensitivity" = "red", "Specificity" = "blue")) +
ylim(0, 1) +
labs(
x = "Device",
y = "Value",
title = "Sensitivity and Specificity by Device"
) +
theme_minimal()
```

Device-specific HSROC/SROC plots with 95% confidence ellipses

```
# Saved as Windows Metafile (.wmf)
# -------------------------------
library(mada)
# Split data by device
data_apple <- subset(data, Device == "Apple")
data_samsung <- subset(data, Device == "Samsung")
data_withings <- subset(data, Device == "Withings")
# Fit Reitsma models
model_apple <- reitsma(data_apple)
model_samsung <- reitsma(data_samsung)
model_withings <- reitsma(data_withings)
# Save as Windows Metafile
win.metafile("device_hsroc_curves.wmf", width = 10, height = 4)
```

```
# Three plots side-by-side
par(mfrow = c(1, 3), mar = c(4, 4, 3, 1))
# Apple
plot(model_apple, main = "Apple", level = 0.95, sroclwd = 2)
points(
1 - (data_apple$TN / (data_apple$TN + data_apple$FP)),
data_apple$TP / (data_apple$TP + data_apple$FN),
pch = 16,
col = "blue"
)
# Samsung
plot(model_samsung, main = "Samsung", level = 0.95, sroclwd = 2)
points(
1 - (data_samsung$TN / (data_samsung$TN + data_samsung$FP)),
data_samsung$TP / (data_samsung$TP + data_samsung$FN),
pch = 16,
col = "blue"
)
# Withings
plot(model_withings, main = "Withings", level = 0.95, sroclwd = 2)
points(
1 - (data_withings$TN / (data_withings$TN + data_withings$FP)),
data_withings$TP / (data_withings$TP + data_withings$FN),
pch = 16,
col = "blue"
)
# Close file
dev.off()
getwd()
```





Script 10: Data Handling Subgroup Analysis

Key Results:

- Worst-case sensitivity ≈ 0.783
- Discarded sensitivity ≈ 0.961

Interpretation: Excluding poor-quality data inflates accuracy estimates; worst-case results likely better reflect real-world performance.

```
# Worst-case vs discarded data handling analysis
# -------------------------------
# Load libraries
library(dplyr)
library(meta)
library(mada)
library(ggplot2)
# -------------------------------
# 1. Create grouping variable
# -------------------------------
data_grouped <- data %>%
  mutate(
    Handling = case_when(
      StudyID %in% c("Racine2022", "Briosa2025", "Mannhardt2023",
"AbuAlrub2022") ~ "Worst-case",
      StudyID %in% c("Muller2024", "Wouters2025") ~ "Discarded",
      TRUE ~ NA_character_
    )
  )
# Exclude studies with unclear handling, e.g. OscaAsensi2021
data_grouped_clean <- data_grouped %>%
  filter(!is.na(Handling))
# Check included studies
table(data_grouped_clean$Handling)
data_grouped_clean[, c("StudyID", "Device", "Handling")]
```

```
StudyID   Device   Handling
1   AbuAlrub2022    Apple Worst-case
2   AbuAlrub2022  Samsung Worst-case
3   AbuAlrub2022 Withings Worst-case
4  Mannhardt2023    Apple Worst-case
5  Mannhardt2023   Fitbit Worst-case
6  Mannhardt2023  Samsung Worst-case
7  Mannhardt2023 Withings Worst-case
8     Muller2024    Apple  Discarded
9     Briosa2025    Apple Worst-case
```

```
10   Wouters2025    Apple  Discarded
11    Racine2022    Apple Worst-case
```

```r
# -------------------------------
# 2. Sensitivity meta-analysis by handling method
# -------------------------------
sens_handling <- metaprop(
  event = TP,
  n = TP + FN,
  studlab = StudyID,
  data = data_grouped_clean,
  sm = "PLOGIT",
  method.tau = "ML",
  subgroup = Handling
)
summary(sens_handling)
```

```
proportion           95%-CI   Handling
AbuAlrub2022      0.8700 [0.7880; 0.9289] Worst-case
AbuAlrub2022      0.8800 [0.7998; 0.9364] Worst-case
AbuAlrub2022      0.7800 [0.6861; 0.8567] Worst-case
Mannhardt2023     0.8525 [0.7383; 0.9302] Worst-case
Mannhardt2023     0.8525 [0.7383; 0.9302] Worst-case
Mannhardt2023     0.5738 [0.4406; 0.6996] Worst-case
Mannhardt2023     0.6557 [0.5231; 0.7727] Worst-case
Muller2024        0.7222 [0.4652; 0.9031]  Discarded
Briosa2025        0.6907 [0.6205; 0.7550] Worst-case
Wouters2025       1.0000 [0.8843; 1.0000]  Discarded
Racine2022        0.7792 [0.7054; 0.8420] Worst-case

Number of studies: k = 11
Number of observations: o = 940
Number of events: e = 729

                     proportion           95%-CI
Common effect model      0.7755 [0.7477; 0.8011]
Random effects model     0.8015 [0.7261; 0.8602]

Quantifying heterogeneity (with 95%-CIs):
 tau^2 = 0.3825; tau = 0.6184
 I^2 = 74.6% [54.1%; 86.0%]; H = 1.99 [1.48; 2.67]

Test of heterogeneity:
          Q d.f.  p-value
 Wald 39.44   10 < 0.0001
 LRT  57.35   10 < 0.0001
```

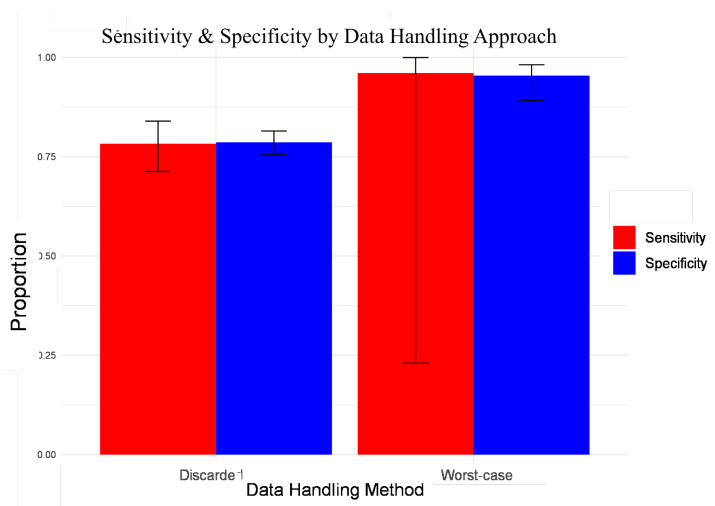
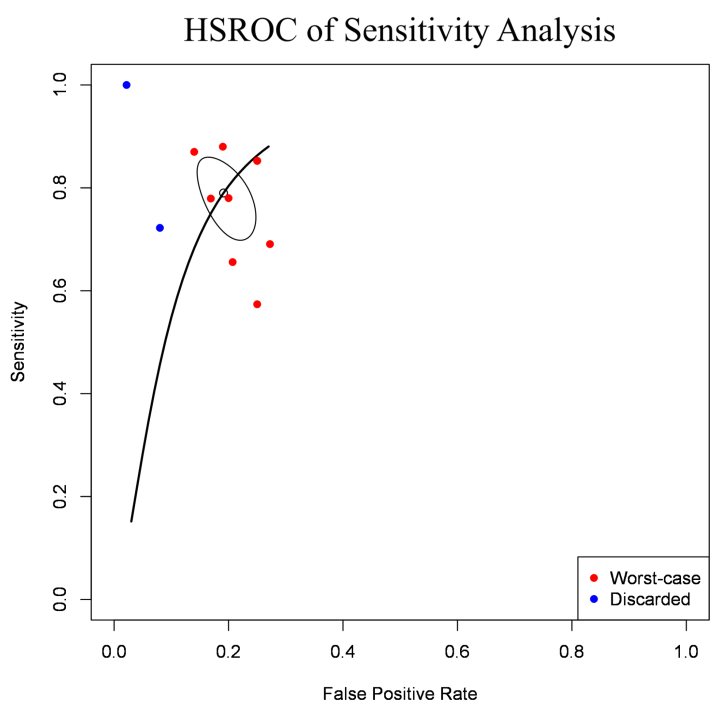
```
Results for subgroups (common effect model):
                         k proportion           95%-CI     Q   I^2
Handling = Worst-case    9     0.7691 [0.7403; 0.7956] 39.33 79.7%
Handling = Discarded     2     0.8958 [0.7731; 0.9560]  0.00  0.0%

Test for subgroup differences (common effect model):
                  Q d.f. p-value
Between groups 3.92    1  0.0477

Results for subgroups (random effects model):
                         k proportion           95%-CI  tau^2    tau
Handling = Worst-case    9     0.7827 [0.7127; 0.8395] 0.2518 0.5018
Handling = Discarded     2     0.9606 [0.2296; 0.9995] 5.5676 2.3596

Test for subgroup differences (random effects model):
                  Q d.f. p-value
Between groups 0.72    1  0.3963

Details of meta-analysis methods:
- Random intercept logistic regression model
- Maximum-likelihood estimator for tau^2
- Calculation of I^2 based on Q
- Logit transformation
- Clopper-Pearson confidence interval for individual studies
```

```r
# -------------------------------
# 3. Specificity meta-analysis by handling method
# -------------------------------
spec_handling <- metaprop(
  event = TN,
  n = TN + FP,
  studlab = StudyID,
  data = data_grouped_clean,
  sm = "PLOGIT",
  method.tau = "ML",
  subgroup = Handling
)
summary(spec_handling)
```

```
 proportion           95%-CI   Handling
AbuAlrub2022      0.8600 [0.7763; 0.9213] Worst-case
AbuAlrub2022      0.8100 [0.7193; 0.8816] Worst-case
AbuAlrub2022      0.8000 [0.7082; 0.8733] Worst-case
Mannhardt2023     0.7500 [0.6698; 0.8193] Worst-case
Mannhardt2023     0.7500 [0.6698; 0.8193] Worst-case
Mannhardt2023     0.7500 [0.6698; 0.8193] Worst-case
Mannhardt2023     0.7929 [0.7162; 0.8567] Worst-case
```

```
Muller2024        0.9200 [0.8340; 0.9701]  Discarded
Briosa2025        0.7276 [0.6725; 0.7780] Worst-case
Wouters2025       0.9783 [0.9237; 0.9974]  Discarded
Racine2022        0.8310 [0.7980; 0.8606] Worst-case

Number of studies: k = 11
Number of observations: o = 1897
Number of events: e = 1525

                     proportion           95%-CI
Common effect model      0.8039 [0.7854; 0.8212]
Random effects model     0.8239 [0.7677; 0.8687]

Quantifying heterogeneity (with 95%-CIs):
 tau^2 = 0.2708; tau = 0.5204
 I^2 = 74.2% [53.0%; 85.8%]; H = 1.97 [1.46; 2.65]

Test of heterogeneity:
          Q d.f.  p-value
 Wald 38.70   10 < 0.0001
 LRT  56.63   10 < 0.0001

Results for subgroups (common effect model):
                        k proportion           95%-CI     Q   I^2
Handling = Worst-case   9     0.7896 [0.7697; 0.8082] 19.68 59.4%
Handling = Discarded    2     0.9521 [0.9071; 0.9759]  2.69 62.8%

Test for subgroup differences (common effect model):
                   Q d.f.  p-value
Between groups 20.62    1 < 0.0001

Results for subgroups (random effects model):
                        k proportion           95%-CI  tau^2    tau
Handling = Worst-case   9     0.7864 [0.7552; 0.8145] 0.0349 0.1869
Handling = Discarded    2     0.9540 [0.8914; 0.9813] 0.1476 0.3842

Test for subgroup differences (random effects model):
                   Q d.f. p-value
Between groups 12.89    1  0.0003

Details of meta-analysis methods:
- Random intercept logistic regression model
- Maximum-likelihood estimator for tau^2
- Calculation of I^2 based on Q
- Logit transformation
- Clopper-Pearson confidence interval for individual studies
```

```
# -------------------------------
# 4. Extract pooled subgroup estimates
# -------------------------------
logit_to_prob <- function(x) exp(x) / (1 + exp(x))
handling_results <- data.frame(
  Handling = levels(factor(data_grouped_clean$Handling)),
  Sensitivity = logit_to_prob(sens_handling$TE.random.w),
  Sens_Lower = logit_to_prob(sens_handling$lower.random.w),
  Sens_Upper = logit_to_prob(sens_handling$upper.random.w),
  Specificity = logit_to_prob(spec_handling$TE.random.w),
  Spec_Lower = logit_to_prob(spec_handling$lower.random.w),
  Spec_Upper = logit_to_prob(spec_handling$upper.random.w)
)
print(handling_results)

  Handling Sensitivity Sens_Lower Sens_Upper Specificity
Worst-case  Discarded   0.7826703  0.7126676  0.8394601   0.7863592
Discarded  Worst-case   0.9606192  0.2296292  0.9994993   0.9539970
           Spec_Lower Spec_Upper
Worst-case  0.7552254  0.8145056
Discarded   0.8913670  0.9812775

# -------------------------------
# 5. Bar graph with 95% CI
# -------------------------------
plot_data <- data.frame(
  Handling = rep(handling_results$Handling, 2),
  Metric = rep(c("Sensitivity", "Specificity"), each = nrow(handling_results)),
  Value = c(handling_results$Sensitivity, handling_results$Specificity),
  Lower = c(handling_results$Sens_Lower, handling_results$Spec_Lower),
  Upper = c(handling_results$Sens_Upper, handling_results$Spec_Upper)
)
ggplot(plot_data, aes(x = Handling, y = Value, fill = Metric)) +
  geom_bar(stat = "identity", position = position_dodge(width = 0.9)) +
  geom_errorbar(
    aes(ymin = Lower, ymax = Upper),
    position = position_dodge(width = 0.9),
    width = 0.2
  ) +
  scale_fill_manual(values = c("Sensitivity" = "red", "Specificity" = "blue"))
+
  ylim(0, 1) +
  labs(
    x = "Data handling method",
    y = "Value",
    title = "Sensitivity and Specificity by Data Handling Method"
  ) +
```

```
  theme_minimal()
win.metafile("handling_bar_plot.wmf", width = 7, height = 5)
ggplot(plot_data, aes(x = Handling, y = Value, fill = Metric)) +
  geom_bar(stat = "identity", position = position_dodge(width = 0.9)) +
  geom_errorbar(
    aes(ymin = Lower, ymax = Upper),
    position = position_dodge(width = 0.9),
    width = 0.2
  ) +
  scale_fill_manual(values = c("Sensitivity" = "red", "Specificity" = "blue"))
+
  ylim(0, 1) +
  labs(
    x = "Data handling method",
    y = "Value",
    title = "Sensitivity and Specificity by Data Handling Method"
  ) +
  theme_minimal()
dev.off()

# -------------------------------
# HSROC / SROC curve with study points
# Save directly to file
# -------------------------------

library(mada)
library(devEMF)

handling_model <- reitsma(data_grouped_clean)

# Make sure FPR and Sensitivity columns exist
data_grouped_clean$FPR <- data_grouped_clean$FP /
  (data_grouped_clean$FP + data_grouped_clean$TN)

data_grouped_clean$Sensitivity <- data_grouped_clean$TP /
  (data_grouped_clean$TP + data_grouped_clean$FN)

dir.create("Outputs/Figures", recursive = TRUE, showWarnings = FALSE)

emf("Outputs/Figures/handling_hsroc_plot.emf", width = 7, height = 7)

par(mar = c(5, 5, 4, 2))

plot(
  handling_model,
  sroclwd = 2,
  main = "HSROC Curve by Data Handling Method"
```

```
)

points(
  data_grouped_clean$FPR[data_grouped_clean$Handling == "Worst-case"],
  data_grouped_clean$Sensitivity[data_grouped_clean$Handling == "Worst-case"],
  pch = 16,
  col = "red"
)

points(
  data_grouped_clean$FPR[data_grouped_clean$Handling == "Discarded"],
  data_grouped_clean$Sensitivity[data_grouped_clean$Handling == "Discarded"],
  pch = 16,
  col = "blue"
)

legend(
  "bottomright",
  legend = c("Worst-case", "Discarded"),
  col = c("red", "blue"),
  pch = 16
)

dev.off()
```